北京郵電大學

Queen Mary University of London

Joint Programme Assessments 2022/23

Answer Sheet

(To be used in conjunction with the provided alternative assessment questions and the supplementary instructions)

Complete the information below about yourself very carefully.

QM student number	200978947
BUPT student number	2020213248
Class number	2020215107
Module code	EBU6018
Word Count	1758

I hereby confirm that the submitted work is entirely my own and I have not (i) used the services of any agency or person(s) providing specimen, model or ghostwritten work in the preparation of the work I submit for this open book assessment; (ii) given assistance in accessing this paper or in providing specimen, model or ghostwritten work to other candidates submitting for this open-book assessment.

Signature *Tangxun Wu*

For examiners ONLY

Do not write in this section

	Q1	Q2	Q3	Q4			Sum
A							
B							
C							
D							
E							
F							
TOTAL							

Question	1	Do not write in this column
Section(s)	a	

Question 1

a)

My BUPT ID is EVEN:

Therefore:

$$A=\begin{bmatrix}2 & -1 & -2\\ -1 & y & 4\\ 2 & z & 4\end{bmatrix}$$

put $v_1=\begin{bmatrix}2\\-1\\2\end{bmatrix}$ $v_2=\begin{bmatrix}x\\y\\z\end{bmatrix}$ $v_3=\begin{bmatrix}-2\\4\\4\end{bmatrix}$

Then $\langle v_1, v_2\rangle = [2,-1,2]\begin{bmatrix}x\\y\\z\end{bmatrix} = 2x-y+2z$

$\langle v_2, v_3\rangle = [x,y,z]\begin{bmatrix}-2\\4\\4\end{bmatrix} = -2x+4y+4z$

$\because$ basis is orthogonal

$\therefore \begin{cases}2x-y+2z=0\\-x+2y+2z=0\end{cases} \Rightarrow \begin{cases}3y+6z=0\\2x+4z=0\end{cases} \Rightarrow \begin{cases}x=-2z\\y=-2z\\z=z\end{cases}$

put $z=1$,

then $x=-2$, $y=-2$

so $v_2=\begin{bmatrix}-2\\-2\\1\end{bmatrix}$

---

b)

i)

performing STFT:

Step 1: choose an appropriate window of finite length.

Step 2: place the window on top of the signal (at the beginning of the signal) at t = 0.

Step 3: Truncate the signal via using the chosen window.

Step 4: Apply FT to the portion of signal within the window.

Step 5: Incrementally shift the window to the right, so as to investigate the frequency component of this signal at different time locations.

Step 6: repeat from step 3.

Question	1	Do not write in this column
Section(s)	b	

b)

ii)

Shape:

Shape influences a lot about the final result.

For example, if the window is rectangular, there will be discontinuities, which will lead to distortion of the spectrum.

Also, it requires to be translated to avoid overlap at the edges or gaps.

Therefore, a rectangular window is a poor choice.

A better shape of window can be “bell” shapes, like Hamming, Hann, Gaussian, etc., because the energy of this kind of window is concentrated near the middle, which allows the window to overlap, as well as minimise distortion.

Besides, this kind of window can also help invert the transform.

Width:

Width also influences a lot about the final result.

For example, narrow window means signal within the window can be stationary, so the time resolution is good, but has poor localisation in the frequency domain.

Wide window is the opposite, that is, meaning poor time resolution but good frequency resolution, because we will have many cycles of them to be able to calculate the frequencies.

Therefore, the window width has a trade-off between time resolution and frequency resolution, which is reflected by the Uncertainty Principle.

Question	2	Do not write in this column
Section(s)	a	

Question 2

a)

i)

Advantages:

1. because the DCT basis functions are orthogonal, the transform is reversible.

2. the implementation of this transform is fast. Because compared with DFT, it assumes signal is even, real and periodic, so it just uses cosines for calculation and doesn't apply any complex arithmetic. Obviously, its output is real, too.

3. when analysing the DCT output, most energy will concentrate in just a few coefficients, so we can achieve compression about the data by discarding some unnecessary coefficients while keeping most of the original signal.

ii)

Separable transform: we call a transform separable if this transform has more than one dimension, and each dimension can do transform individually.

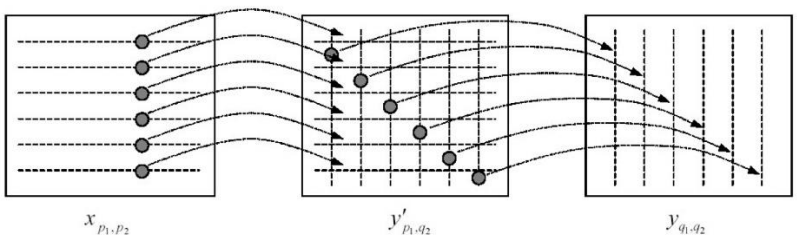
For DCT of a 2D image, we can firstly let each row of the 2D image to be transformed as just 1 dimension.

After that, on the bases of this 1D transform, we apply DCT again in each column of this 1D transform. In this case, we complete 2D transform of this image.

We can use a diagram to indicate this:

Separable Transforms

May be implemented by applying the one dimensional transform first to the rows of the image and then to its columns (note that changing the application order does not change the result).



Question	2	Do not write in this column
Section(s)	a	

iii)

My BUPT ID is EVEN:

Therefore:

s[n] = [3, 2, 7, -4]

$$DCT = \langle S, \Psi \rangle = \begin{bmatrix} 0.50 & 0.50 & 0.50 & 0.50 \\ 0.65 & 0.27 & -0.27 & -0.65 \\ 0.50 & -0.50 & -0.50 & 0.50 \\ 0.27 & -0.65 & 0.65 & -0.27 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 7 \\ -4 \end{bmatrix} = \begin{bmatrix} 4 \\ 3.2 \\ -5 \\ 5.14 \end{bmatrix}$$

So y[n] = [4, 3.2, -5, 5.14]

iv)

At first, the image will be sub-divided into 8*8 blocks of data.

Then, we will apply DCT to the image by multiplying the 8*8 DCT matrix with an 8 points input sequence, so as to get the result of an 8 points output sequence. After that, we repeat this step, until all blocks in the image data have done DCT.

In this way, we can calculate the correlation between the input data and the cosine waves at different frequencies.

Then, we will set a threshold on DCT output, which means setting outputs below the threshold to 0, so that the compression is done.

Besides, when using DCT for compression, if we inverse the transform, we can get an accurate approximation of the original image. The data we discard cannot be perceived through human eyes, actually.

Therefore, DCT, which transform the data from spatial domain to frequency domain, is an effective way to help compression in images.

Question	2	Do not write in this column
Section(s)	b	

Question 2

b)

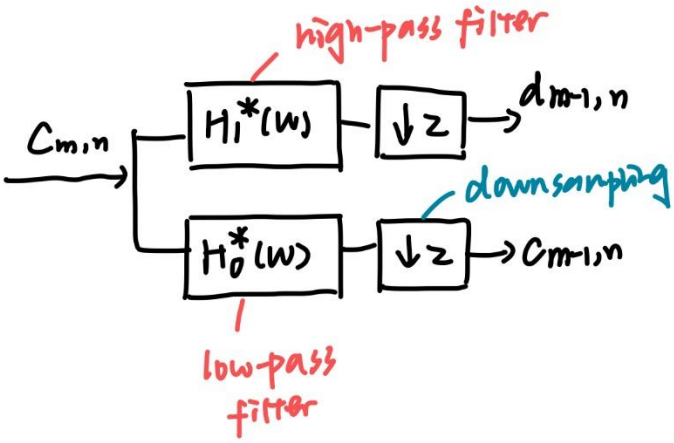
Filtering: in the equation, h0 represents low pass filter to obtain the next level of coarse detail, while h1 represents high pass filter to obtain the next level of fine detail.

Downsampling: in the equation, we can see step of 1 in n can be reflected to steps of 2 in i. This shows that downsampling is carried out by letting the number of values at each level become half compared to the previous stage. In other words, from n to i, downsampling is executed by a factor of 2.

Therefore, the concept of an analysis filterbank is that:

when setting c(m,i) as the scaling function coefficients at level m (the coarse detail), we can obtain the scaling function coefficients c(m-1,n) and wavelet series coefficients d(m-1,n) at level m-1 by recursively using this equation.

The diagram of the filterbank:

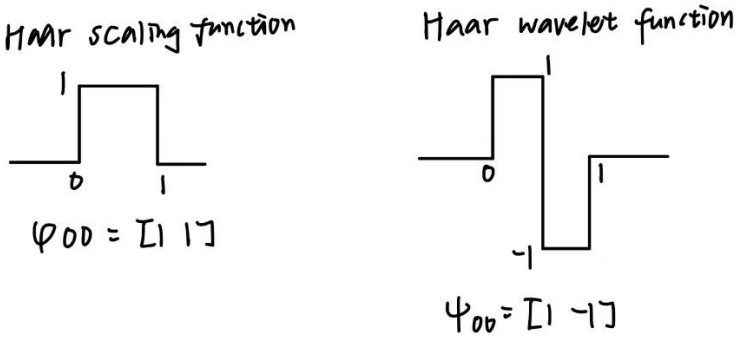


Question	3	Do not write in this column
Section(s)	i	

Question 3

i)

1)



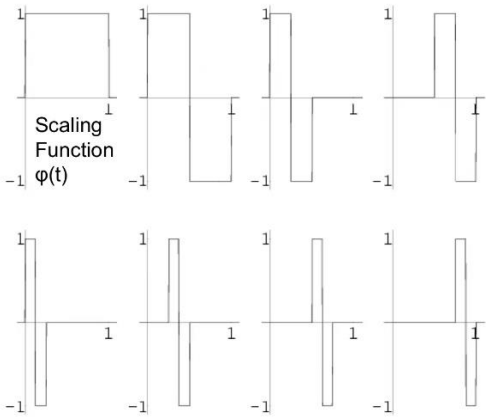
2)

The definition of wavelet function is shown below, we know that:

$$\psi(x) = \begin{cases} 1 & 0 \le x \le \frac{1}{2} \\ -1 & \frac{1}{2} \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\psi_{jk}(x) = \psi(2^j x - k)$$

j for scaling, k for translation



Through haar functions' diagram (on top right):

we can easily see that they satisfy wavelet functions' equation.

To look the diagram at the details,

not only the haar function has the short duration, but also it is oscillatory.

Also, it forms a haar function family (packet) by scaling and translation, so that it can be used to perform a number of transforms.

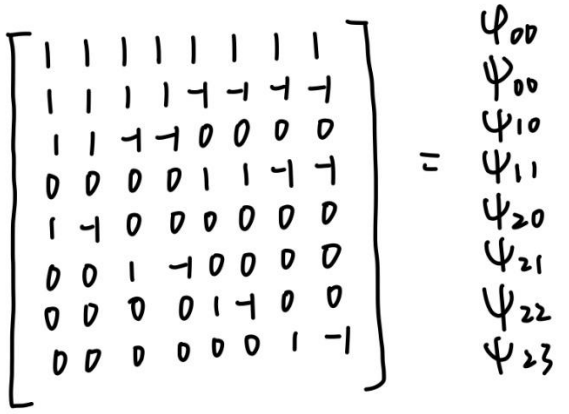
Therefore, the haar Function can be used as a simple wavelet, because it satisfies all the property of the wavelet funtion.

Besides, because haar functions are just defined in the time interval [0,1], and all of them are rectangular-shaped, orthogonal, and real numbers, they are the simplest type of wavelet function.

Question	3	Do not write in this column
Section(s)	ii	

ii)

1) the 8*8 unnormalized Haar transform matrix:



2)

The first row has identical value, and can be used to measure the average value, by dividing the output by sqrt(N).

The second row measures a low frequency component, which means slow changing value.

Similar to DCT, the following rows correspond to increasing frequencies:

For the third row and forth row, they measure the moderate frequency component.

For the rows from fifth to eighth, they measure the high frequency components, which just have a narrow shape but a rapid change in the diagram.

iii)

$S[k]$ = [9.5, −10.21, 1.01, 14.2, 2.09, −0.08, 0.19, −2.89]

1. The first coefficient is 9.5, indicating the average value of input sequence is 3.36.

2. The second coefficient is large, meaning that the input has lots of low-frequency components during the time.

3. The fourth coefficient is the largest, showing that the input has a fast-changing value (sharpest decrease) during that period t = [0,1/2]. Also, the frequency represented by the fourth coefficient can be the most common and largest part of frequency in the signal.

4. For the last four coefficients, we know that the input has little high frequency components during the total time.

Question	4	Do not write in this column
Section(s)	a	

Question 4

a)

i)

The process of doing data compression is that:

Statistically decoupling the multidimensional data by computing covariance matrix.

Then the eigenvector of this matrix is used as basis vectors for transforming the data.

After that, discard the transformed data on the least principal components (least variance). We can choose to set last column of basis vector to 0, or the last row of transformed data to 0 to achieve compression.

Therefore, KLT can preserve all subspace that has largest variance, so as to achieve optimum energy concentration and coding gain.

ii)

Advantages: it can maximise the coding gain and get highest quality data, i.e., maximise the SNR after a specific level of compression.

The reason for this is that it keeps the subspace that has largest variance, and just discard the data on the least principal components, which are not important.

Disadvantages:

Firstly, the calculation is complex.

Because we need to have solution of eigenvector decomposition, which is computationally intensive. Also, the forward and inverse transform need big calculation capacity.

Secondly, KLT is not suitable for non-linear datasets.

Because orthogonal principal components fit the nonlinearity poorly, leading to large weights for each principal component. So the compression by KLT is lossy.

Finally, KLT is not suitable for low-dimensional datasets.

Because sometimes the dataset just has 2 principal components, so it’s difficult to compression data without losing much.

Question	4	Do not write in this column
Section(s)	b	

Question 4

b)

i)

Uncertainty principle states: the product of the uncertainty in time and the uncertainty in frequency is a constant.

The equation:

$$\Delta t \Delta w \geq \frac{1}{2}$$

The delta t is the uncertainty in time.

The delta w is the uncertainty in frequency.

ii)

it means that we can't localise a function in time and frequency simultaneously.

(Know the time and frequency information accurately at the same time).

If delta t is small, then delta w is large.

If delta t is large, then delta w is small.

If the signal is fixed in time, it has infinite bandwidth.

If the signal is fixed in frequency, it has infinite duration.

We can't know exactly what frequency components exist at a particular point in time, we just know a specific time interval and corresponding frequency bands.

If we want to get best time-frequency resolution based on uncertainty principle, we should apply Gaussian function.

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Section(s)		

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